

Control work 5 · Inequalities, modulus and accuracy

The Chapter II assessment, then a work-on-mistakes lesson sorted by error type.

LESSON 60–61

2 × 40 MIN

ALGEBRA 8, §§15–19

STAGE 9 · 3.4, 4.3

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess inequalities in one unknown, systems, modulus, error and bounds.
- Sort every lost mark into one of four error types.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§15–19
Cambridge	Learner’s Book pp. 74–81, 96–102
Workbook	Workbook 3.4, 4.3

01

Explanation

Lesson 60 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- tasks 1–2 · solve a linear inequality, one with a negative coefficient (§15)
- task 3 · solve a system and write the interval (§16)
- task 4 · a double inequality (§16)
- task 5 · a modulus equation (§17)
- task 6 · a modulus inequality (§17)
- task 7 · bounds or relative error (§§18–19)

Lesson 61 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

- The sign not reversed after dividing by a negative number.
- Only one case taken for a modulus equation — the negative case forgotten.
- The two modulus inequality shapes confused — writing two rays where one interval belongs, or the reverse.

4. **Bracket types mixed up** when writing an interval, or ∞ given a square bracket.

02 Worked examples

EXAMPLE 1 Find the error: $|x - 2| = 5$, so $x = 7$

- ① Only the positive case was taken.
This is error 2.

- ② $x - 2 = 5$ or $x - 2 = -5$
Both cases.

- ③ $x = 7$ or $x = -3$

ANSWER

$$x = 7 \text{ or } x = -3$$

EXAMPLE 2 Find the error: $|x| > 3$, so $-3 < x < 3$

- ① That is the answer for $|x| < 3$, not for $>$.
The two shapes are swapped.

- ② “More than 3 from zero” is everything **outside**.

- ③ $x < -3$ or $x > 3$

ANSWER

$$(-\infty; -3) \cup (3; +\infty)$$

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $-3x < 12 \Rightarrow x < -4$

① Dividing by

-3 — a negative number.

② The sign turns:

$x > -4$

③ Test

$x = 0$:

$0 < 12$ ✓, and

$0 > -4$ ✓.

One test settles it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Solution set	Yechimlar to'plami	Множество решений
Interval	Oraliq	Промежуток
Modulus	Modul	Модуль
Bound	Chegara	Граница
Error	Xato	Ошибка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$-3x < 12$ gives:

$$x < -4$$

$$x > -4$$

$$x < 4$$

$$x > 4$$

$|x - 2| = 5$ gives:

$$x = 7$$

$$x = -3$$

$$x = 7 \text{ or } x = -3$$

no solutions

$|x| > 3$ gives:

$$-3 < x < 3$$

$$x < -3 \text{ or } x > 3$$

$$x > 3$$

$$x < 3$$

$x \geq 2$ as an interval is:

$(2; +\infty)$

$[2; +\infty)$

$(2; +\infty]$

$[2; +\infty]$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Solve $4x - 5 > 7$

ANSWER $x > 3$

2. **Task 2.** Solve $-2x + 1 \geq 9$

ANSWER $x \leq -4$

3. **Task 3.** Solve: $x > 2$ and $x \leq 6$

ANSWER $(2; 6]$

4. **Task 4.** Solve $-3 < 2x + 1 \leq 7$

ANSWER $(-2; 3]$

5. **Task 5.** Solve $|x - 1| = 4$

ANSWER $x = 5$ or $x = -3$

6. **Task 6.** Solve $|x| \leq 5$

ANSWER $[-5; 5]$

7. **Task 7.** 8 m to the nearest metre. Give the bounds.

ANSWER $7.5 \leq \ell < 8.5$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Solve $5x + 2 \leq 17$

ANSWER $x \leq 3$

2. **Task 2.** Solve $3 - 4x > 15$

ANSWER $x < -3$

3. **Task 3.** Solve: $x \geq -1$ and $x < 4$

ANSWER $[-1; 4)$

4. **Task 4.** Solve $-6 \leq 3x < 9$

ANSWER $[-2; 3)$

5. **Task 5.** Solve $|2x + 1| = 9$

ANSWER $x = 4$ or $x = -5$

6. **Task 6.** Solve $|x - 2| > 3$

ANSWER $x < -1$ or $x > 5$

7. **Task 7.** Exact 40, approximation 39. Find the relative error.

ANSWER 2.5%

Hard · stretch and reason

7 PROBLEMS

1. Find the error: $-3x < 12$, so $x < -4$

ANSWER The sign was not reversed. Correct: $x > -4$.

2. Find the error: $|x - 2| = 5$, so $x = 7$

ANSWER Only one case. Correct: $x = 7$ or $x = -3$.

3. Find the error: $|x| > 3$, so $-3 < x < 3$

ANSWER The shapes are swapped. Correct: $x < -3$ or $x > 3$.

4. Find the error: $x \geq 2$ written as $(2; +\infty]$

ANSWER Correct: $[2; +\infty)$.

5. Find the error: $|x + 4| = -1$, so $x = -5$ or $x = -3$

ANSWER A modulus is never negative — there are no solutions.

6. Find the error: “7 cm to the nearest cm lies between 6.9 and 7.1”

ANSWER The bounds are 6.5 and 7.5 — half a unit, not a tenth.

7. Find the error: “ $x > 5$ and $x < 2$ gives $(5; 2)$ ”

ANSWER The sets do not overlap; there are no solutions.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quadratic equations begin next lesson.

- Solve $-5x + 3 \geq 18$
- Solve $|3x - 2| = 10$
- Solve $|x + 1| < 6$
- A length reads 9.4 cm to 1 d.p. Give the bounds.
- Write out the four errors from this lesson in your own words with an example each.